

Mean-field Dynamics

Setup: let $\mathcal{M}$ be a smooth Riemannian manifold, Haar msr. $\mathfrak{h}$
 $\mathcal{P}(\mathcal{M})$ measures on $\mathcal{M}$, $\mathcal{P}_{ac} = \{\mu \in \mathcal{P}(\mathcal{M}) : \mu \ll \mathfrak{h}\}$

Particles undergo motion in a drift field:

$$v : t, \mu_t \mapsto \text{vector field on } \mathcal{M}$$

abuse notation: $v(t, X, \mu) \in T_X(\mathcal{M})$

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Fix $\mu_0 \in \mathcal{P}(\mathcal{M})$, $X_0 \sim \mu_0$, and drift $v(\cdot)$. Dynamics:

$$\partial_t X_t = v(t, X_t, \mu_t) \quad (\text{LAG})$$

$$\partial_t \mu_t = \operatorname{div}_{\mathcal{M}}(v(t, \cdot, \mu_t) \mu_t) \quad (\text{EUL})$$

Upshot: $X_t \sim \mu_t$ for all $t \geq 0$.